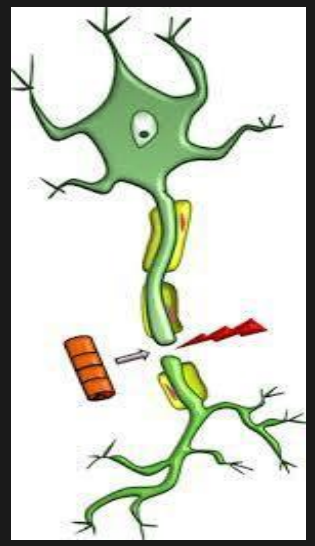


Data interpretation and assessment

Fundamental of Neurosciences (FON) – PAD1

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By the end of this tutorial

1. Know the assessment structure

Explain the ICA and examination formats, timing and permitted resources.

2. Choose the right graph

Select appropriate graph types for common biomedical data.

3. Build clearer graphs

Use Excel to create readable, well-labelled scientific graphs.

4. Interpret data

Move from observation to evidence, comparison, explanation and conclusion.

5. Answer MCQ/SAQ well

Approach traditional and non-traditional assessment questions strategically.

6. Avoid common errors

Reduce preventable mistakes with units, scales, uploads and timing.

Assessment overview

In-course assessment (ICA)

50% of the module

50 marks.
VLE-based data interpretation assessment using experimental datasets.

Format: Table conversion, Graph generation and Multiple choice question

Operational arrangement: 90 minutes, closed-book, Excel permitted for graphing,

Exam condition: no generative AI, no websites, no personal notes.

Written unseen examination

50% of the module.

One-hour written unseen exam with MCQs and SAQs. Tests conceptual knowledge, mechanisms and application across the module.

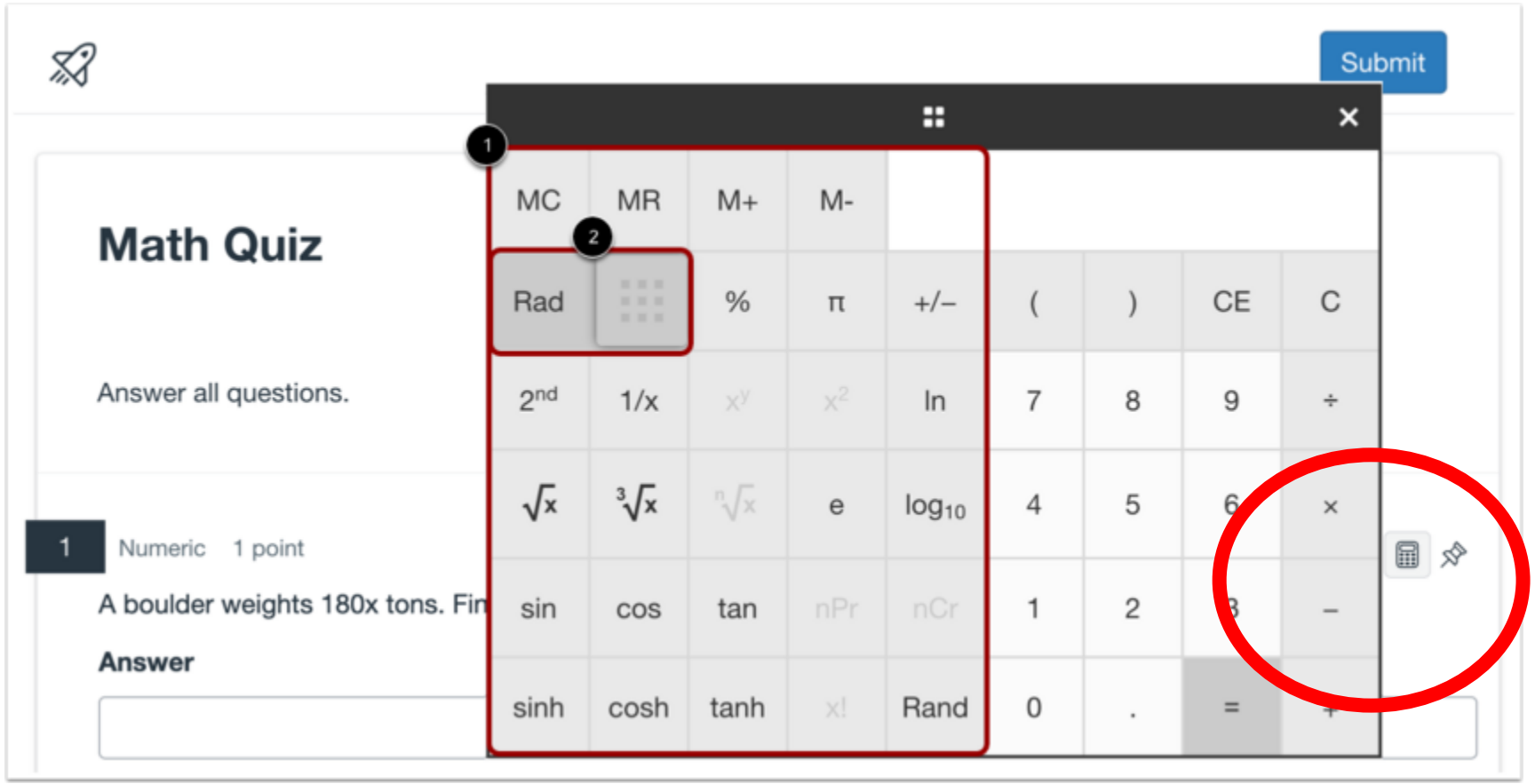
the ICA is not a memory test of the practical sheet;

it is a test of using experimental information, data handling and scientific reasoning.

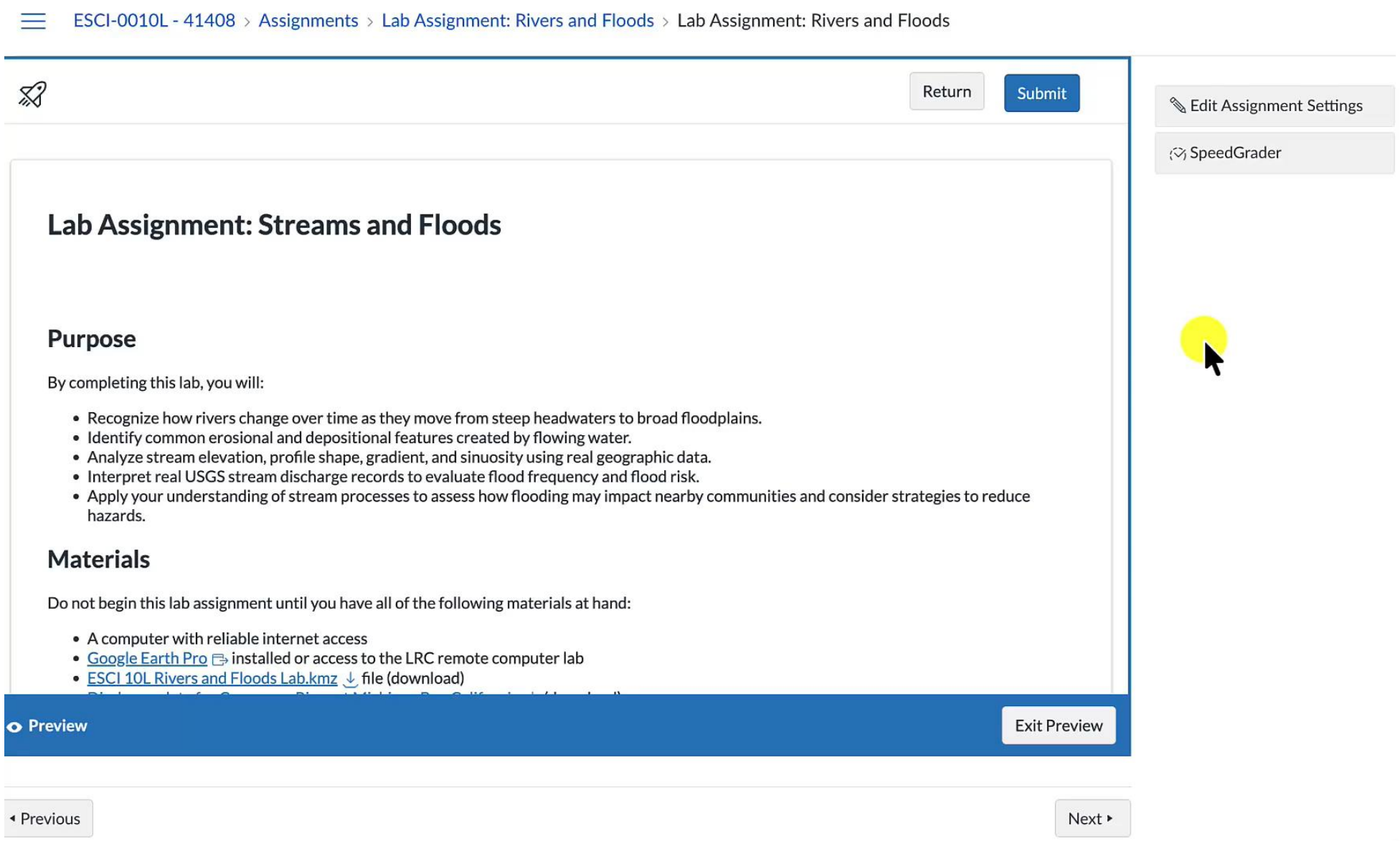
ICA operational instructions

- The ICA contains two experimental datasets and is worth 50 marks.
- Duration: 90 minutes. It will automatically submit when time expires.
- It is individual and closed-book. Excel and the computer calculator are permitted.
- Generative AI, websites, messaging apps, personal notes and another person's work are not permitted.
- There are two graphing questions. Each requires one .xlsx upload, so two .xlsx files are submitted in total.
- Each Canvas question is marked independently; a mistake in one question does not automatically reduce later marks.

How do I use Calculator on Canvas?



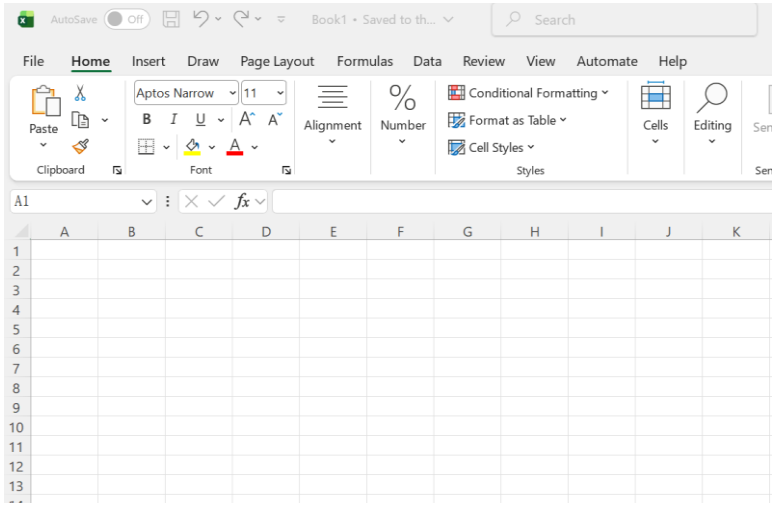
How do I insert a table using the Rich Content Editor?



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How do I insert an Excel sheet on Canvas?

1. Draw your graph in excel and save



2. Upload your graph/excel file on Canvas

Drag and drop here or Browse

Maximum 1 file(s) allowed

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What this tutorial will not do ?

We will practise

Graph selection, Excel habits, interpreting trends, calculations such as percentage change, and question-answering strategy.

We will not provide

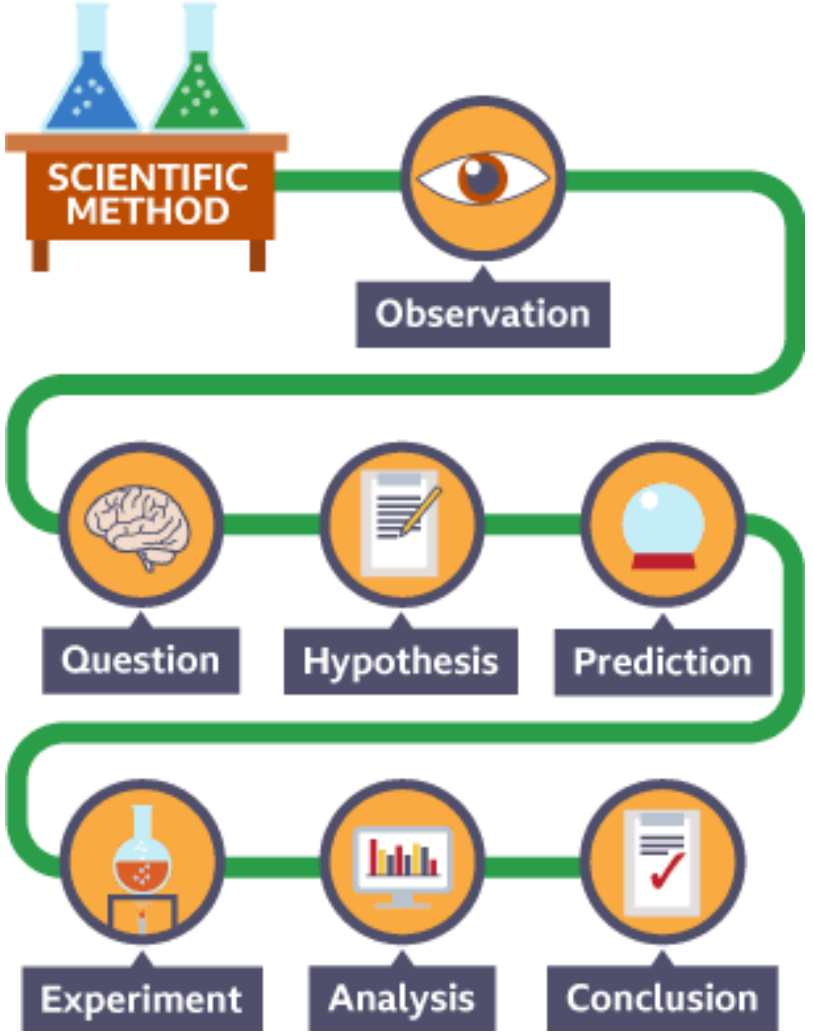
Summative ICA datasets, exact summative questions, correct answers to summative questions, or model responses for the real ICA.

From experiment to answer

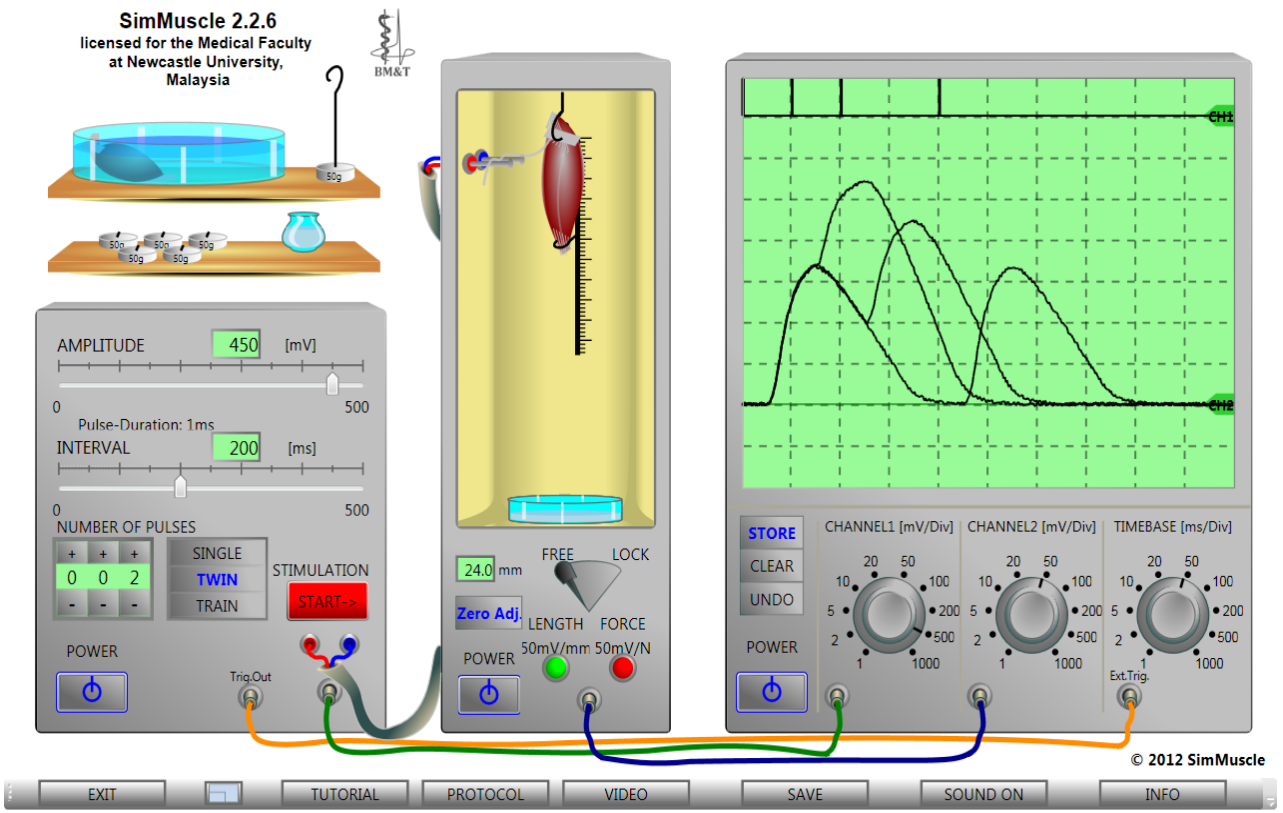
Question → Variables → Raw values → Conversion

↓

Graph → Pattern → Mechanism → Conclusion



SimMuscle practical: why it matters



What the practical trains

Observe muscle response, measure values from a trace, convert units, plot response curves.

What the ICA tests

Can you use an experimental description and a dataset to make an evidence-based interpretation?

- Use the practical to understand variables, axes, thresholds, plateaus and physiological mechanisms.
- If you missed details in the practical, the ICA stimulus should still contain the information needed to attempt the task.

What did we investigate?

One virtual nerve–muscle preparation. Four experimental questions.

01 Stimulus intensity

How does voltage change muscle force?

02 Twin pulses

What happens when stimuli arrive closer together?

03 Stimulation frequency

How do twitches become tetanus?

04 Load & repeated pulses

How does mechanical demand or repetition affect responses?

Before analysing: what were we measuring?

LOCK • ISOMETRIC

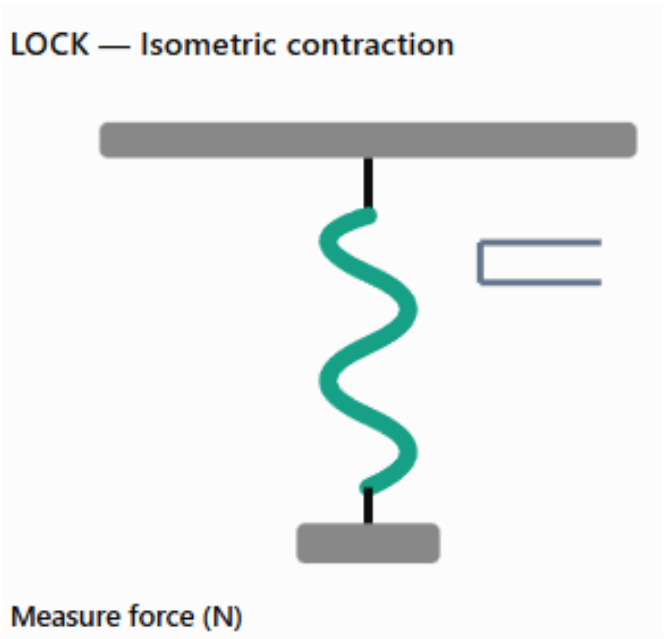
Force changes

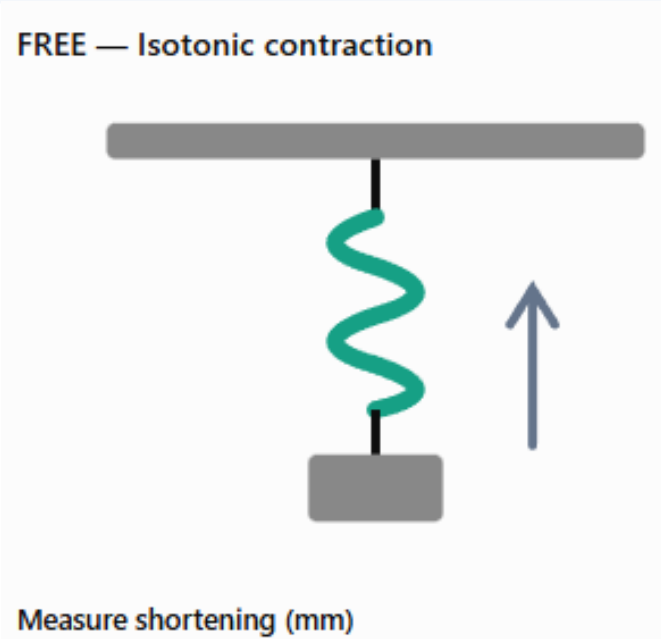
Measure force in newtons (N)

FREE • ISOTONIC

Muscle length changes

Measure shortening in millimetres (mm)





Do not mix up force (N) and shortening (mm): they are different outcome measures.

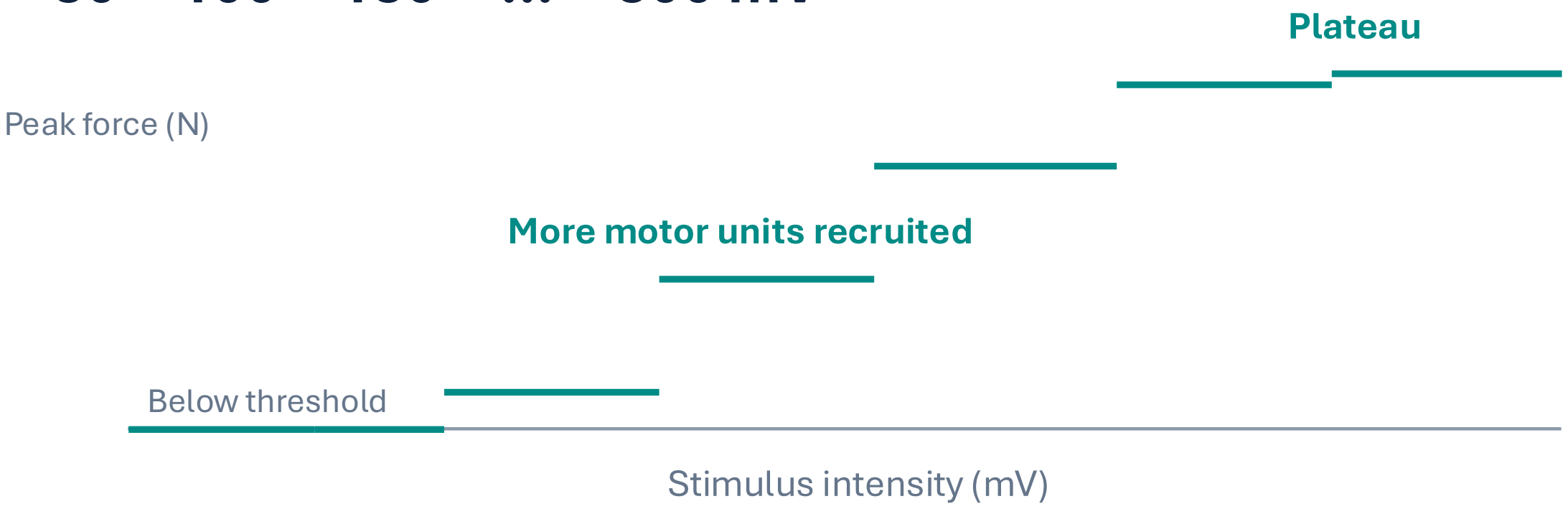
Experiment 1 | Increasing stimulus intensity

CHANGED Stimulus voltage

MEASURED Peak force (N)

MODE LOCK

50 → 100 → 150 → … → 500 mV



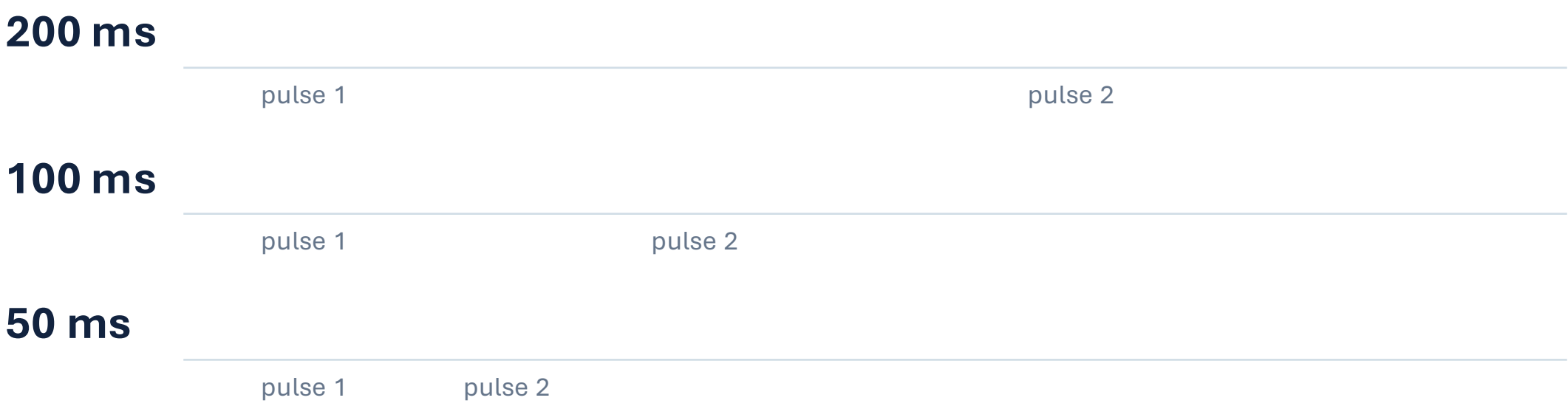
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Experiment 2 | Two stimuli, different intervals

FIXED 450 mV

CHANGED 200 / 100 / 50 ms

MODE FREE



Key idea: The second twitch may overlap with the first before relaxation is complete.

Experiment 3 | From twitches to tetanus

200 ms	**5 Hz**	8 pulses	**Separate twitches**
100 ms	**10 Hz**	16 pulses	**Unfused tetanus**
50 ms	**20 Hz**	32 pulses	**Fused tetanus**

f (Hz) = 1000 ÷ interval (ms) • Repeat all three settings in LOCK and FREE modes.

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Experiment 4 | Load and repeated stimulation

A • INCREASE THE LOAD

0 → 100 → 200 → 300 g

Fixed 400 mV; single pulses

Record response size and duration

B • REPEAT THE PULSES

1 → 2 → 10 → 20 → 30

Fixed 400 mV; 250 ms interval

Observe changes across the trace

Turning practical results into analysis

1	**IDENTIFY**	What did you change? What did you measure?
2	**TABULATE**	Include headings and appropriate units.
3	**PLOT**	Choose sensible axes and chart types.
4	**INTERPRET**	Describe the trend using evidence.
5	**EXPLAIN**	Link the trend to physiological mechanisms.

Step 1: define the scientific question

Question type

Does stimulus intensity affect muscle response? Does load affect shortening? Does frequency affect summation?

Independent variable

What the experimenter changes: voltage, load, time, drug concentration, frequency, genotype or condition.

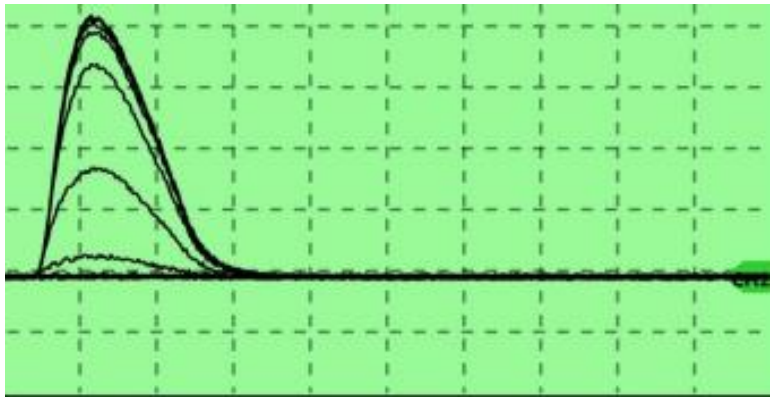
Dependent variable

What is measured: force, contraction amplitude, time to peak, rate, absorbance, fluorescence or cell count.

If you cannot identify the independent and dependent variables, your graph choice and interpretation will probably be weak.

Step 2: check units and calibration

Raw measurement	Calibration	Converted value
2.8 grid squares height	1 square = 1 N	2.8 N
3.1 grid squares duration	1 square = 50 ms	155 ms
1.5 grid squares time to peak	1 square = 50 ms	75 ms



Common mistakes

Mixing up mV, ms, mm and N; forgetting that timebase affects all horizontal measurements; labelling graph axes with raw grid squares rather than converted units.

Best habit

Write the conversion rule before calculating. Then use the same rule consistently throughout the table.

Step 3: organise data before graphing

Stimulus voltage (mV)	Force (N)	Time to peak (ms)
50	0.0	0
100	0.0	0
130	0.6	40
170	1.4	60
220	2.8	75
270	4.1	85

- One row = one observation or condition.
- One column = one variable.
- Use headers with units.
- Avoid merged cells in the data area.
- Keep calculations in separate columns when possible.

How to create a chart or graph in Excel | Microsoft

Microsoft Excel

Create a chart

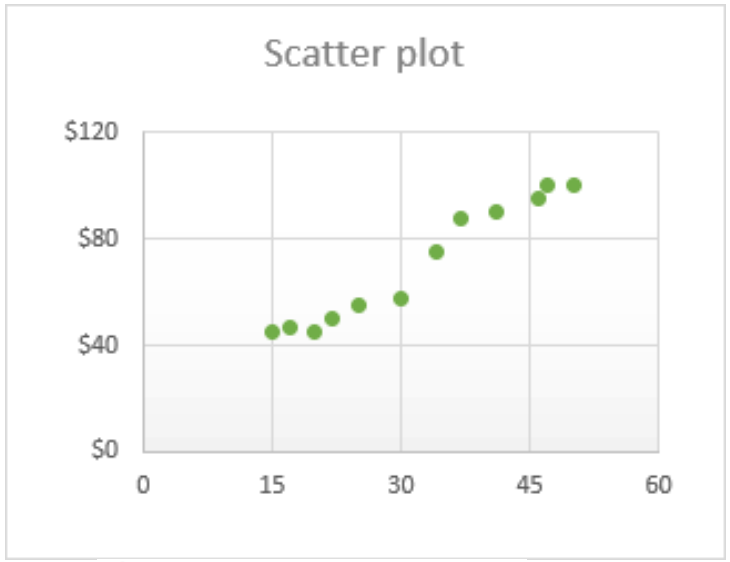


Choosing the graph type

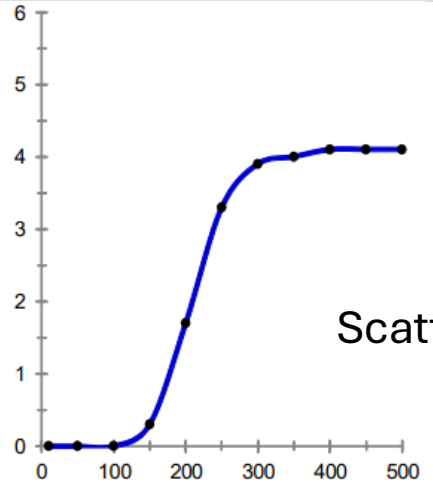
Data / question	Better graph choice	Why
Continuous x and continuous y, e.g. voltage vs force	XY scatter with line/curve	Preserves numeric spacing on the x-axis.
Time course, repeated measurements over time	Line graph	Shows sequence and change over time.
Categories or conditions, e.g. control vs treatment	Bar/column or dot plot	Compares group summaries.
Distribution of many values	Histogram or box plot	Shows spread, skew and outliers.
Many values across two dimensions	Heatmap	Shows patterns across rows and columns.
Means with variability	Dot/bar + error bars	Communicates uncertainty or variation.

Graph type 1: XY scatter

Use XY scatter when x-values are actual numbers, not just category labels.



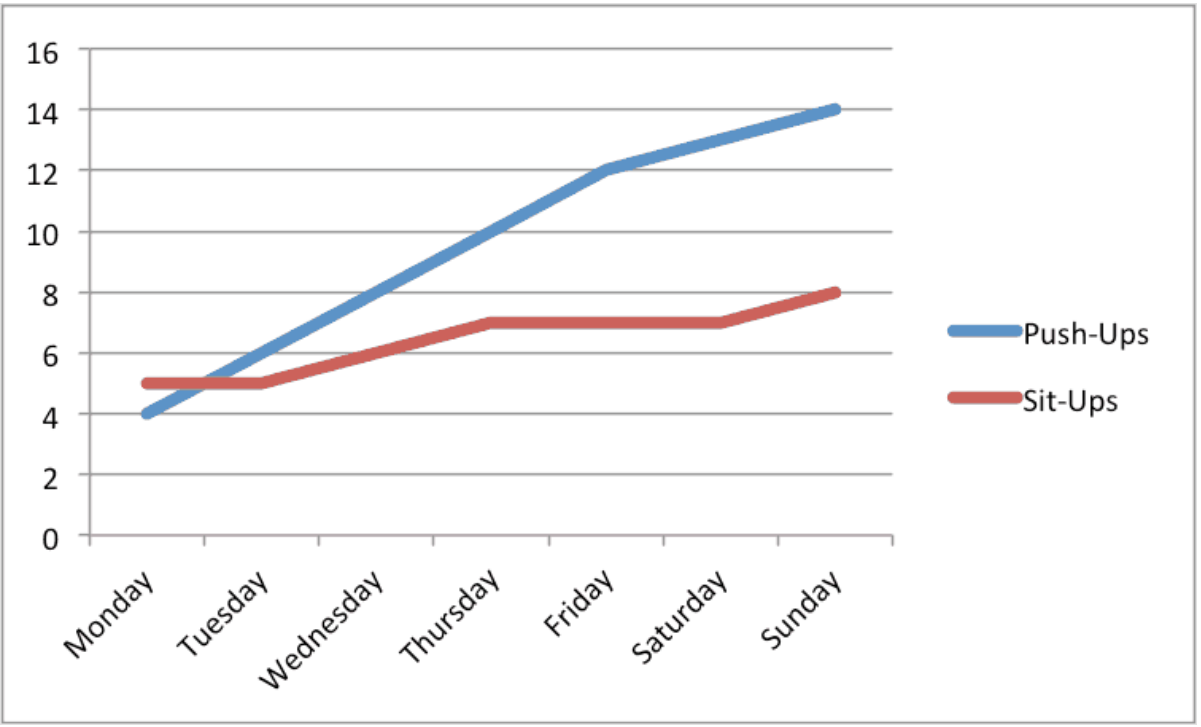
- Good for stimulus–response and dose–response relationships.
- Can reveal thresholds, saturation and non-linear patterns.
- Use a clear x-axis scale; do not treat unevenly spaced numeric values as equal categories.



Scatter with straight lines and markers

Graph type 2: line graph

Line graphs connect values because sequence matters.

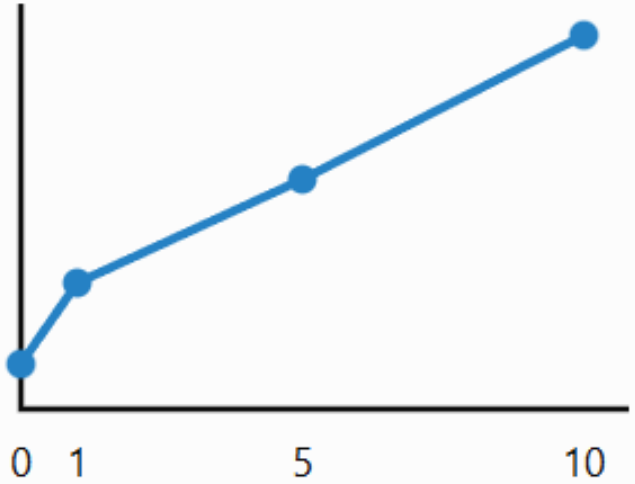


- Use for time courses: calcium signal over time, fatigue across repeated stimuli, concentration after dosing.
- Do not use a line simply to join unrelated categories.
- If categories are independent, a bar chart or dot plot may be clearer.

Example

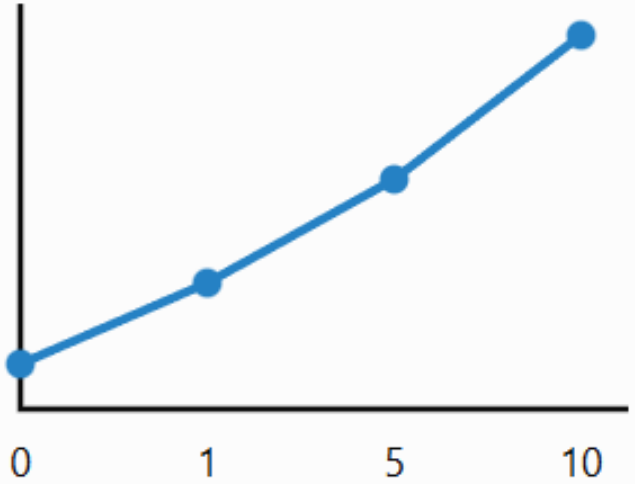
Suppose you measure muscle contraction force at 0, 1, 5 and 10 volts.

XY scatter with lines



Actual numerical spacing

Excel line graph

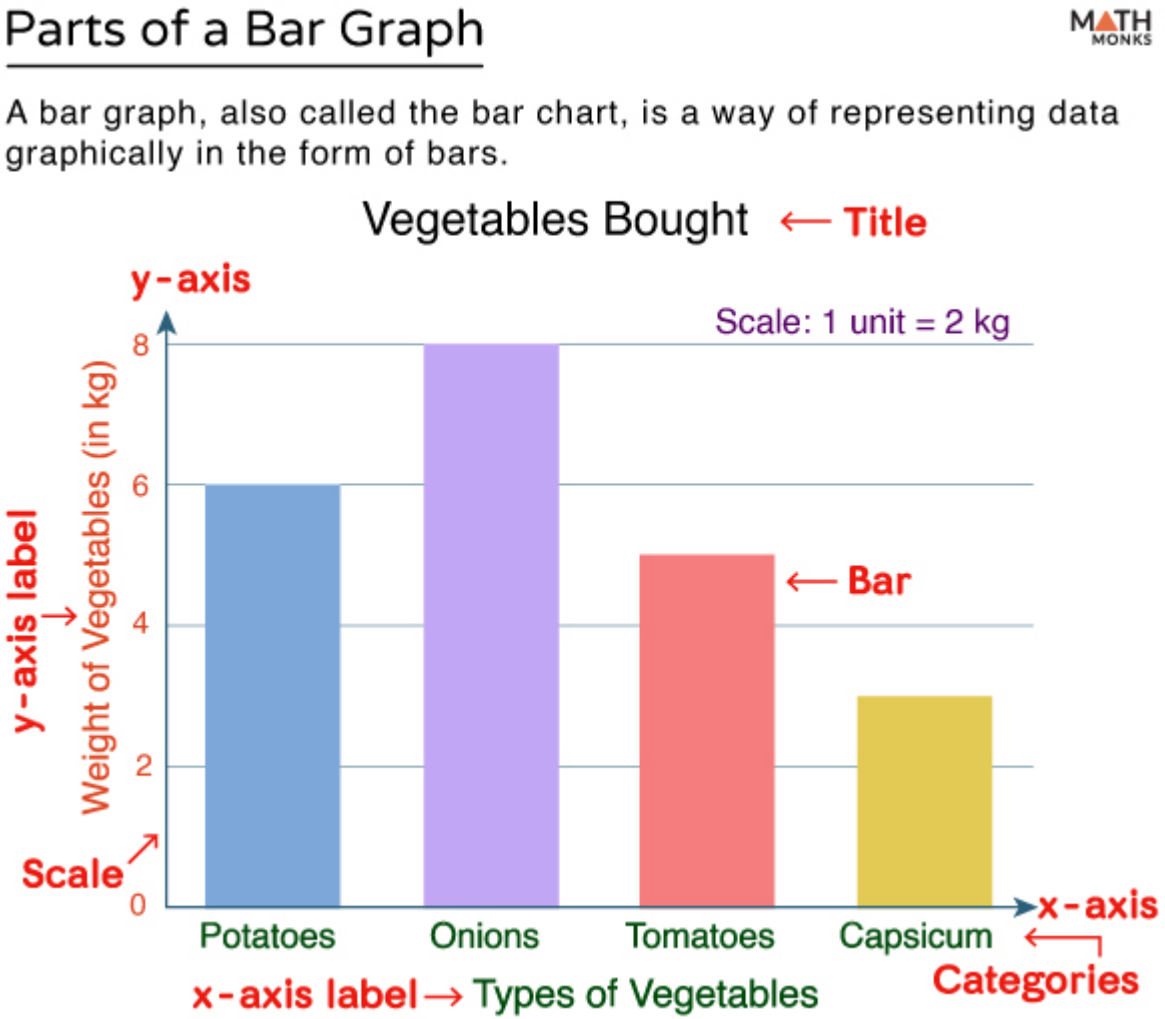


Equal spacing between categories

	XY Scatter with Lines and Markers	Line Graph
X-axis	Numerical (value axis)	Categorical (category axis)
X-values: 0, 1, 5, 10	Spacing reflects the differences: 1, 4, 5	All points equally spaced
Points connected?	Yes	Yes
Suitable for scientific data?	Yes, particularly continuous numerical X-values	Yes, particularly equally spaced time points or ordered categories

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Graph type 3: bar/column chart



When it works

Comparing conditions such as no load, 100 g, 200 g and 300 g; comparing mean responses between treatment groups.

When to be careful

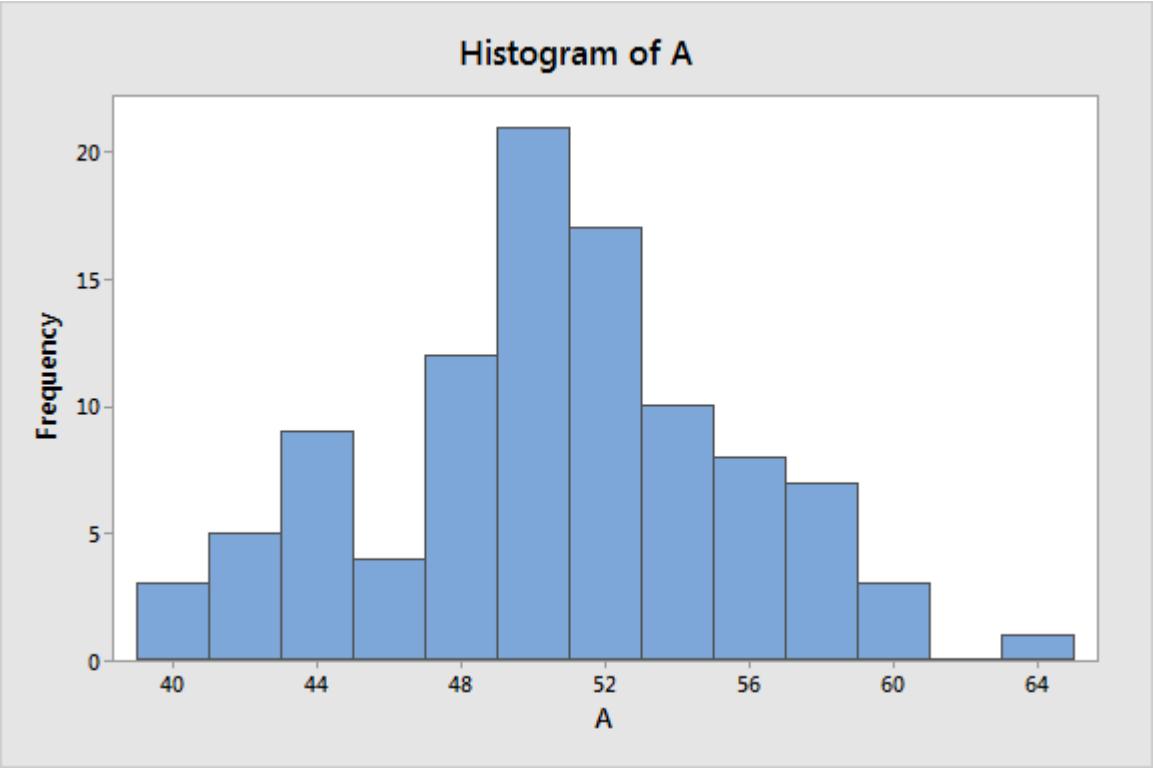
Bar charts hide the distribution. If you have individual data points, consider plotting dots or box plots.

Minimum requirements

Label axes with units, start at zero when comparing magnitudes, and avoid 3D effects.

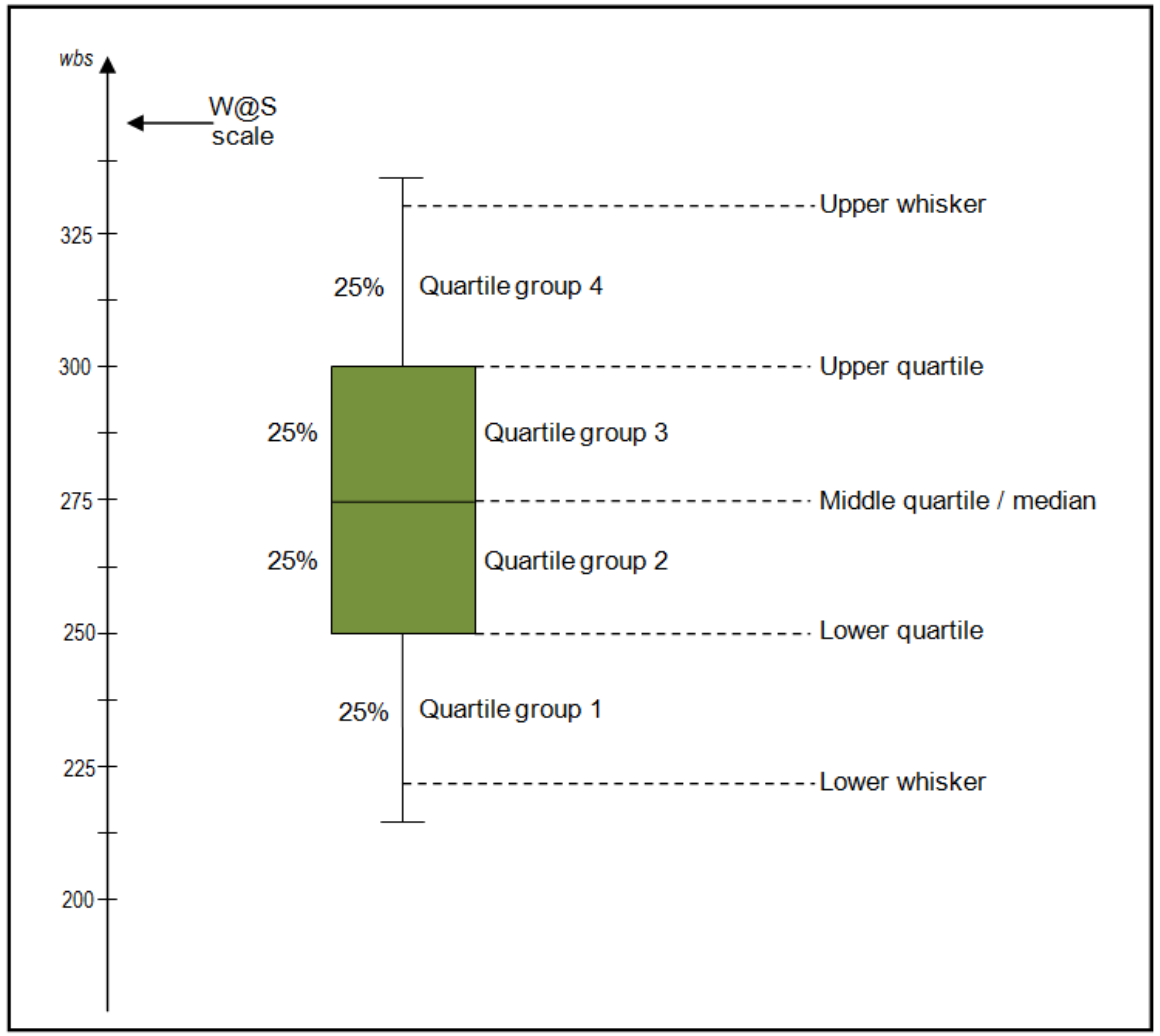
Graph type 4: histogram

A histogram answers: how are many values distributed?



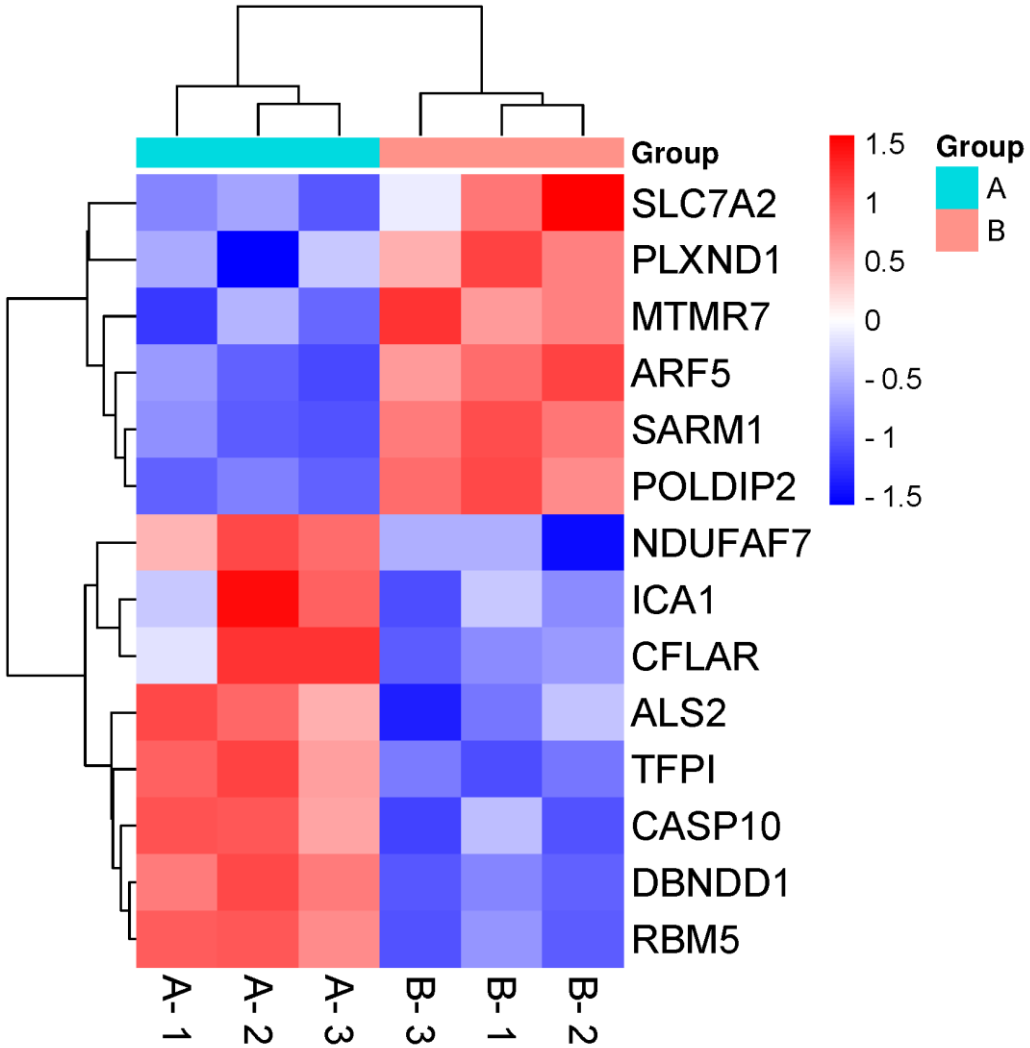
- Use when you have many observations of the same variable.
- Shows centre, spread and skew.
- Not the best choice for four experimental conditions.
- Bins matter: different bin widths can change the visual impression.

Graph type 5: box plot



- Shows median, interquartile range and approximate spread.
- More informative than bars when there are many biological replicates.
- Often used for gene expression, clinical measures, enzyme activity or cell counts.
- A box plot is not needed when you only have one value per condition.

Graph type 6: heatmap



- Common in gene expression, proteomics, cell signalling and imaging summaries.
- Colour shows magnitude; rows and columns create pattern recognition.
- Always include a colour scale in real scientific figures.
- Not suitable when precise numerical values are the main message.

Error bars: what do they mean?

SD

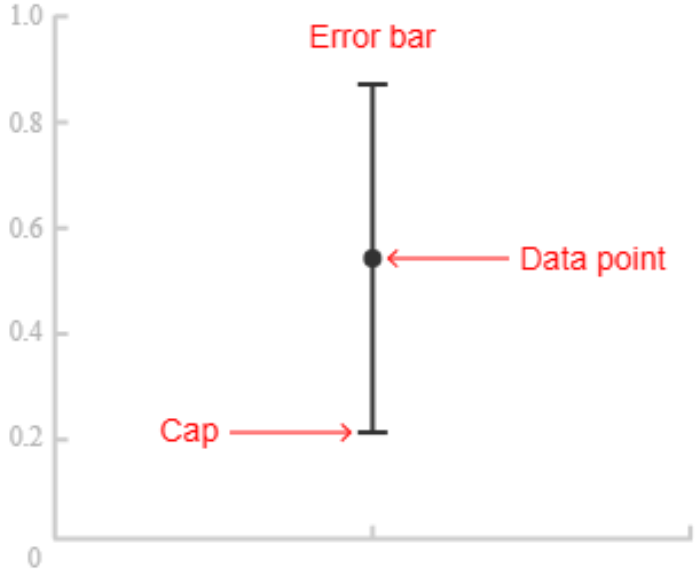
Standard deviation: spread of individual data points.

SEM

Standard error of the mean: precision of the estimated mean.

CI

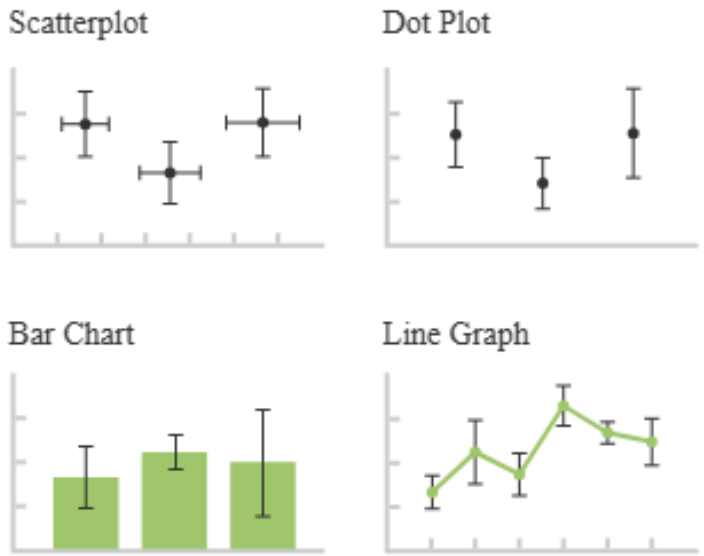
Confidence interval: likely range of the population estimate.



Error bars are not decoration.

They communicate variation or uncertainty.

- If a question gives only one measured value per condition, error bars are usually not available.
- If a question gives repeats, ask what the question requests: mean ± SD, mean ± SEM, or confidence interval.
- Never guess error bars from the shape of a curve.



Excel workflow for scientific graphs

1. Paste/enter data
Prepare data

2. Check units
Prepare data

3. Select x + y columns
Prepare data

4. Insert graph
Prepare data

5. Label axes
Make and submit

6. Check scale
Make and submit

7. Save workbook
Make and submit

8. Upload
Make and submit

Excel: making an XY scatter graph

Place the independent variable in the left column and dependent variable in the right column.
Highlight both columns including headers.
Insert → Scatter → Scatter with markers or smooth lines.
Add axis titles with units.
Check the x-axis uses the real numeric spacing.
Use a title that states the relationship, not just “Graph 1”.

Better title:
“Effect of stimulus voltage on contraction force”

Weak title:
“Force graph”

Excel: bar chart, heatmap and formatting

Bar/column chart

Use Insert → Column or Bar. Best for categories. Keep the y-axis starting at zero for magnitude comparisons. Use simple colours and avoid 3D.

Heatmap

Use a table and conditional formatting. Useful for many values. Make sure the colour scale is explained.

Graph formatting

Axes labelled with units; readable font; clear scale; no unnecessary gridlines; keep one key message per graph.

Scientific graph checklist

Check	Question to ask yourself	Common preventable error
Graph type	Does it match the variable types?	Using a line chart for unrelated categories.
X-axis	Is the independent variable on the x-axis?	Putting the measured response on the x-axis.
Units	Are units included in both axis labels?	Force without N; time without ms.
Scale	Does the scale show the data honestly?	Cropping the axis to exaggerate a small difference.
Title	Does the title state the relationship?	Title says only “Graph 1”.
File	Is the correct workbook uploaded to the correct question?	Uploading the wrong file or no file.

Data interpretation framework

Trend

What changes?

Evidence

Which values prove it?

Compare

How large is the difference?

Explain

What mechanism could account for it?

Conclude

What can and cannot be concluded?

Template: “As X increases, Y changes from ___ to ___. This suggests ___ because ___.”

Common biomedical data patterns

Pattern	What it means visually	Biomedical examples
Threshold	No response until a minimum stimulus is reached	Action potential initiation, muscle contraction, receptor activation.
Plateau / saturation	Response stops increasing despite more stimulus	Maximum motor unit recruitment, receptor saturation, enzyme Vmax.
Linear increase	Response increases in proportion to input	Calibration curves within a limited range.
Biphasic response	Increase followed by decrease	Optimal preload, dose toxicity, stress responses.
Inverse relationship	One variable increases while another falls	Load vs shortening, drug dose vs cell viability.
Lag / latency	Delay before response appears	Synaptic delay, excitation-contraction coupling, signalling cascade.

Calculation skills that support interpretation

Calculation	Formula / idea	Interpretation use
Percentage change	(new − old) / old × 100	Quantifies how much a response increases or decreases.
Rate	change / time	Compares speed of response under different conditions.
Normalisation	value / maximum or value / control	Compares relative responses when absolute values differ.
Threshold	first non-zero or specified response	Identifies the minimum effective stimulus.
Maximum / plateau	first point where no further meaningful increase occurs	Identifies saturation or recruitment limit.

From data to physiology

Description

What the data show. Example: force increases from 0.6 N to 4.9 N as voltage rises.

Explanation

A plausible mechanism. Example: increasing voltage recruits more motor units.

Speculation

A claim not directly supported by the dataset. Example: identifying a precise molecular defect without evidence.

Good data interpretation says what the evidence supports

Avoids claiming more than the data can show.

Practical-linked example: twin-pulse summation

Exp 2. Superposition of single twitches (twin pulses):

Record the muscle contraction in response to twin pulses (supramaximal amplitude) of different pulse intervals (see table) and determine at what interaval the maximum amplitude is reached. Use the SAVE mode and copy the screen when required.

Directives:

- Select the muscle (right one) from petridish
- Amplitude = 450 mv
- Ch1=500 ; Ch2=50 (mv/Div); Time base (ms/Div)=50
- Number of pulses: Select, **Twin.**
- Interval: a) 200 b)100 c) 50 ms and see the impact of each of them one by one. Save individually or as combined figure.
- Keep the toggle Free (ie:Isotonic); Keep Zero Adj in active mode (ie: blue colour)
- **Each square ≈ 1 unit (Y axis ie: muscle force)**
- **Each square ≈ 50 ms (X axis ie: duration)**

Observation

As the interval between pulses decreases, the muscle has less time to relax fully before the second stimulus.

Data idea

Compare contraction amplitude at different intervals; look for the largest response and explain why it occurs.

Mechanism

Summation occurs because mechanical contraction lasts longer than the nerve action potential and calcium handling takes time.

Practical-linked example: tetanus and frequency

Exp 3. Single Twitches vs.Tetanic Contractions (Both for Isometric vs Isotonic contraction)

Task: Stimulate the muscle with supramaximal pulses of different frequencies both for Isometric and Isotonic conditions that lead to three conditions as below:
a) Isolated single twitches
b) Incomplete contraction (Clonus)
c) Complete tetanic contractions.
d) Can use the other muscle now for this experiment (left one, if you wish)

Directives:

- Amplitude 450 mv
- Channel 2 and Time base both are at **200ms/DIV** constant.
- Channel 1 in (mv/Div) in 500.
- Therefore, **each square ≈ 4 units** (Y axis wise)
- Measure the muscle contraction [in unit **N** (isometric) and **mm (**isotonic)]. Please use the same muscle both for isometric and isotonic contraction.
- Set the "INTERVAL" at **200, 100 and 50** ms for **8,16 and 32 Train intervals** [ie; Number of Pulses] respectively and see the effect.
- Remember: LOCK= Isometric; FREE = Isotonic

Note: To display the sequences in a single oscilloscope screen, pulses (n) have been selected at 8, 16 and 32 train interval which is equivalent to frequency at 5, 10 and 20 hz respectively.

- Low frequency: separated single twitches.
- Intermediate frequency: incomplete tetanus; contractions do not fully relax.
- High frequency: complete tetanus; a sustained contraction is produced.
- Interpretation should connect the visual pattern to repeated stimulation and incomplete relaxation.

- **Frequency is a variable; pattern of contraction is the response.**

Practical-linked example: fatigue and load

Exp 4. Impact of load and fatigue on muscle contraction

A) **Effect of Increased load**

Directives:

- Amplitude: Fixed at 400 mv
- Interval: 250 ms
- Keep the toggle lock (So, isometric); Keep Zero Adj in active mode [ie: blue mode]
- Ch 1 : 500 mv/Div; Ch 2: 50mv/Div; Ch 3: 50timebase (ms/Div)
- **Each square ≈ 1mm (Y axis)**
- **Each square ≈ 50 ms (X axis ie: duration)**
- Number of pulses: Single

(Use the right hand muscle on your petridish for this experiment)

Possible data pattern

A small load may increase shortening due to favourable preload; excessive load can then reduce external shortening.

Interpretation habit

Do not assume “more load always means more contraction”. Look for the actual trend and quote values.

Mechanism language

Use cautious language: “may suggest”, “is consistent with”, “could reflect”.

Traditional MCQ strategy

1. Read the stem first

Identify what the question is really asking before looking at options.

2. Predict an answer

Try to answer in your own words, then compare options.

3. Use elimination

Remove options that contradict core physiology or units.

4. Watch qualifiers

Always, never, first, most likely and except change the logic.

5.Check the mechanism

The right answer should fit the causal pathway, not just include familiar words.

MCQ with simple responses vs. statements

1. Which of the following glial cell types is believed to be functionally related to macrophages?

A. astroglia
B. oligodendroglia
C. microglia
D. Schwann cells
E. ependymal cells

Which of the following statements is correct concerning the autonomic nervous system?

A. The sympathetic and parasympathetic nervous systems can bring about antagonistic physiological effects by innervating different tissues
B. The chemical transmitter at the synapse between pre- and post-ganglionic neurons is noradrenaline in the sympathetic and acetylcholine in the parasympathetic system
C. Autonomic efferents innervate only smooth muscle or cardiac muscle
D. All structures innervated by the sympathetic nervous system are also innervated by the parasympathetic nervous system
E. The autonomic nervous system receives sensory information only from sensory receptors in the viscera

Non-traditional MCQ strategy: Data-based MCQ

Question style	How to approach it
Best supported conclusion	Find the values that directly support each option. Eliminate overclaims.
Threshold / plateau	Identify first non-zero response or first maximum response.
Percentage change	Calculate carefully before choosing; do not estimate visually if exact values are available.
Mechanism interpretation	Choose the option that matches both the data and physiology.
Multiple answer / matching	Treat each statement independently; do not assume there is only one true idea unless the question says so.

SAQ strategy for the written exam

For a 4-mark SAQ, think: definition + mechanism + evidence/example + consequence.

Weak answer	Improved answer
“The muscle contracts more because stimulation is higher.”	“Increasing stimulus voltage recruits more motor units. More active muscle fibres generate a larger whole-muscle force until a plateau is reached.”
“The graph goes up then stops.”	“Force rises steeply between the threshold and mid-range voltages, then approaches a plateau, suggesting most recruitable motor units have been activated.”

Activity 1: choose the best graph

Dataset	Best graph?	Reason
Stimulus voltage and contraction force		
Control, drug A, drug B mean force		
Fluorescence intensity measured every 10 seconds		
Expression of 20 genes across 4 treatments		
100 cell sizes measured from one image		

Discuss, then we will check answers together.

Activity 2: write a three-sentence interpretation

Stimulus (mV)	Force (N)
50	0.0
100	0.0
150	0.3
200	1.7
250	3.3
300	3.9
400	4.1

Task

Write three sentences:
1. Describe the trend.
2. Quote two values as evidence.
3. Give a plausible physiological explanation.

A strong answer describes, quantifies and explains — in that order.

Time management during the ICA

Time	Action
0–5 min	Read instructions and identify the two datasets and two uploads.
5–25 min	Complete first conversion/calculation section.
25–40 min	Create first graph, save workbook and upload.
40–60 min	Complete second conversion/calculation section.
60–75 min	Create second graph, save workbook and upload.
75–88 min	Complete MCQs and check numeric answers.
88–90 min	Confirm both .xlsx files are attached before auto-submit.

Submission and technical checklist

- ✓ Use clear file names, for example: studentID_exp1_graph.xlsx and studentID_exp2_graph.xlsx.
- ✓ Save the workbook before uploading.
- ✓ Check that the workbook contains the required graph(s).
- ✓ Upload the correct .xlsx file to the correct Canvas question.
- ✓ Do not upload screenshots, PDFs or blank workbooks unless specifically instructed.
- ✓ If something goes wrong, inform an invigilator immediately.

What to practise after today

Exam preparation

Practise traditional MCQs and SAQs: definitions, mechanisms and concise explanations.

ICA preparation

Practise data conversion, graphing, non-traditional MCQs and evidence-based interpretation.

Excel preparation

Practise creating, saving and uploading graphs using a clean workbook.

You do not need to memorise one dataset. You need to practise a reliable process.

Final take-home messages

- ❑ Read the assessment instructions before starting the data work.
- ❑ Choose graph type from variable type and scientific question.
- ❑ Use converted units, not raw grid-square labels, unless asked otherwise.
- ❑ Interpret data using Trend → Evidence → Compare → Explain → Conclude.
- ❑ Do not overclaim: say what the data support.
- ❑ Leave time to save and upload both Excel workbooks.

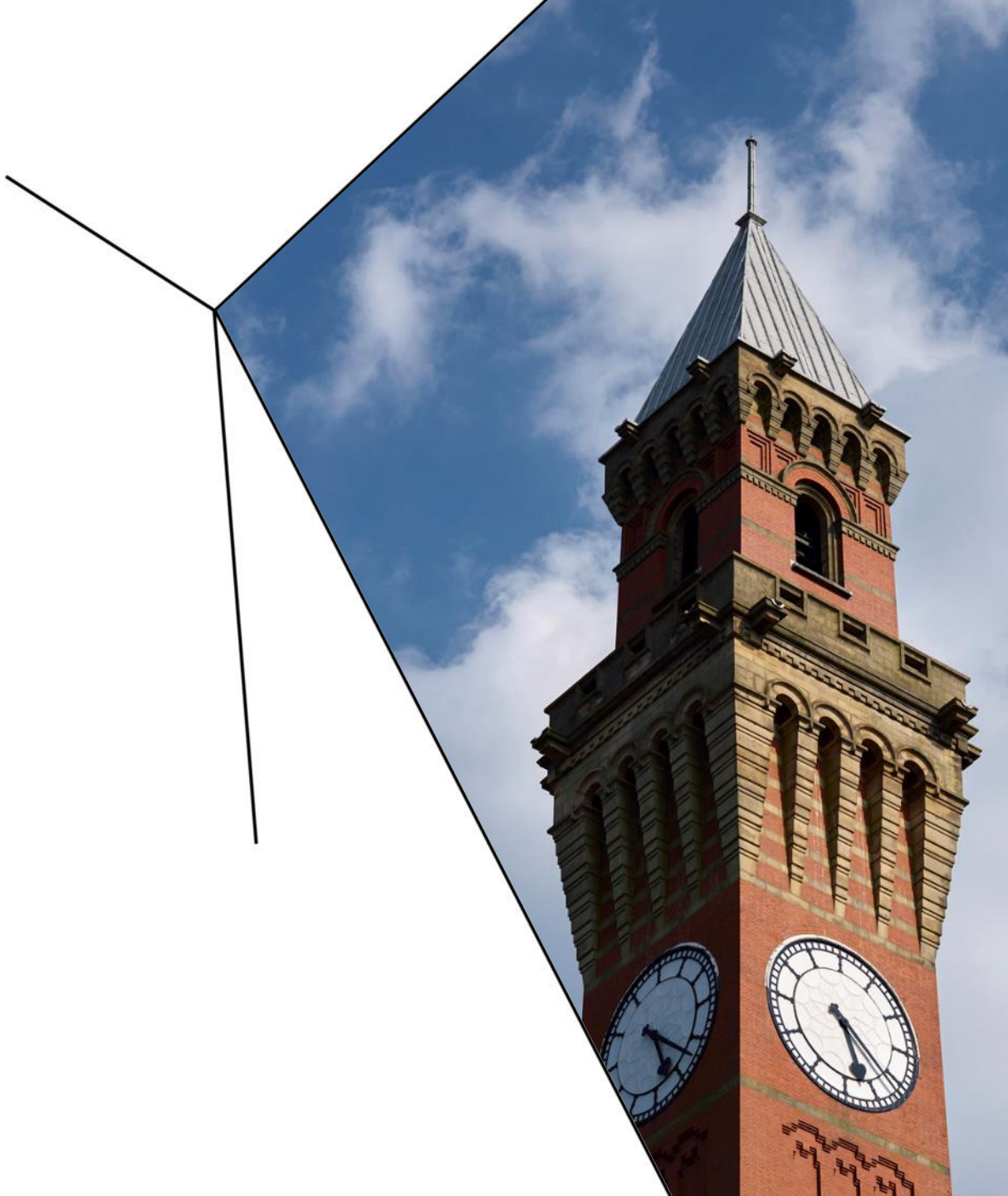
Questions

Thank you

If you have any questions

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